

# Econ 695 - Midterm 1 Solutions

October 8, 2025

**Date and Time: 8:00-9:15 AM, October 8th, 2025**

**Instructions: Write legibly, and explain as carefully as you can what you are doing so we can award partial credit!**

## Question 1 (25 points; 15 minutes)

Consider the distribution of individual incomes,  $y_i$  in a population that is divided into two groups, indicated by the random variable  $D_i \in \{0, 1\}$ . The population regression of interest is

$$y_i = \beta_0 + \beta_1 D_i + u_i \quad (1)$$

which satisfies  $E[u_i] = E[u_i D_i] = 0$ . Suppose that  $Pr(D_i = 1) = p$ .

**(a) (10 points)** Find  $E[y_i]$  - the population mean of  $y_i$  as a function of  $\beta_0, \beta_1$ .

• **Solution:**

- (5 points; no proof needed) In the population regression,  $\beta_0 = E[y_i | D_i = 0]$  and  $\beta_1 = E[y_i | D_i = 1] - E[y_i | D_i = 0]$ .
- (5 points) By the law of iterated expectations,

$$\begin{aligned} E[y_i] &= Pr(D_i = 1)E[y_i | D_i = 1] + (1 - Pr(D_i = 1))E[y_i | D_i = 0] \\ &= Pr(D_i = 1)(\beta_0 + \beta_1) + (1 - Pr(D_i = 1))\beta_0 \\ &= \beta_0 + \beta_1 Pr(D_i = 1) \end{aligned}$$

note: give full credit if the student says the regression with a constant can fit the mean, and therefore  $E[y_i] = \beta_0 + \beta_1 E[D_i]$ .

**(b) (15 points)** Suppose  $D_i = 1$  if person  $i$  is female and 0 otherwise. And we also have  $X_i = 1$  if  $i$  has at least a college degree, and 0 otherwise. What is the conditional expectation function  $E[y_i | D_i, X_i]$ ? Propose a population regression that coincides with the CEF and explain. *Hint: carefully define the regressors based on binary  $D_i$  and binary  $X_i$ .*

• Solution:

- (6 points if the student points out this partition) Given two binary variables, the population can be divided into four (mutually exclusive) groups:

$$(D_i, X_i) \in \{(0, 0), (0, 1), (1, 0), (1, 1)\}$$

- (6 points on a correct population regression = CEF) Consider the following population regression:

$$y_i = \beta_0 + \beta_1 D_i + \beta_2 X_i + \beta_3 D_i X_i + \epsilon_i$$

The conditional expectation function (CEF) can be written as:

$$E[y_i|D_i, X_i] = \beta_0 + \beta_1 D_i + \beta_2 X_i + \beta_3 D_i X_i.$$

- \* If the student proposes  $y_i = \beta_0 + \beta_1 D_i + \beta_2 X_i + u_i$  without the interaction, the population regression is not the CEF. Give **3 points**.
- \* Alternatively, the student may define four indicators for the subgroups  $Z_{1i} = 1$  if  $(D_i, X_i) = (0, 0)$ ,  $Z_{2i} = 1$  if  $(D_i, X_i) = (0, 1)$ ,  $Z_{3i} = 1$  if  $(D_i, X_i) = (1, 0)$ , and  $Z_{4i} = 1$  if  $(D_i, X_i) = (1, 1)$ . The correct population regression would be  $y_i = \sum_{k=1}^4 \beta_k Z_{ki} + \epsilon_i$  and the CEF is  $E[y_i|D_i, X_i] = \sum_{k=1}^4 \beta_k Z_{ki}$ , in which  $\beta_k$  is the group-specific mean. **Give full credits**.
- (3 points on explaining why the population regression matches the CEF) The population regression allows us to fit the mean of each subgroup:

$$\beta_0 = E[y_i|(0, 0)]$$

$$\beta_0 + \beta_1 = E[y_i|D_i = 1, X_i = 0]$$

$$\beta_0 + \beta_2 = E[y_i|D_i = 0, X_i = 1]$$

$$\beta_0 + \beta_1 + \beta_2 = E[y_i|D_i = 1, X_i = 1]$$

Also give full credit in the alternative CEF  $E[y_i|D_i, X_i] = \sum_{k=1}^4 \beta_k Z_{ki}$ , if the student points out  $\beta_k$  is the group-specific mean.

## Question 2 (35 points; 25 minutes)

Consider the following long and short regressions in the population:

$$\text{(long)} \quad y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + u_i \tag{2}$$

$$\text{(short)} \quad y_i = \alpha_0 + \alpha_1 x_{1i} + \alpha_2 x_{2i} + \nu_i$$

in which  $(\beta_0, \beta_1, \beta_2, \beta_3)$  in the long regression is the unique solution that solves  $\min_b E[(y_i - b_0 - b_1 x_{1i} - b_2 x_{2i} - b_3 x_{3i})^2]$ , and  $(\alpha_0, \alpha_1, \alpha_2)$  in the short regression is the unique solution that solves  $\min_a E[(y_i - a_0 - a_1 x_{1i} - a_2 x_{2i})^2]$ .

- (a) (10 points) Derive a formula relating  $\alpha_2$  and  $\beta_2$  on  $x_{2i}$ .

• **Solution:**

- (4 points) Consider an auxiliary regression of the omitted variable  $X_3$  on  $(1, X_1, X_2)$ :

$$x_{3i} = \gamma_0 + \gamma_1 x_{1i} + \gamma_2 x_{2i} + \xi_i$$

The residual is orthogonal to the regressors  $(1, x_{1i}, x_{2i})$ :  $E[\xi_i] = E[\xi_i x_{1i}] = E[\xi_i x_{2i}] = 0$ .

- (2 points) Plug the auxiliary regression into the long regression:

$$\begin{aligned} y_i &= \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 (\gamma_0 + \gamma_1 x_{1i} + \gamma_2 x_{2i} + \xi_i) + u_i \\ &= (\beta_0 + \beta_3 \gamma_0) + (\beta_1 + \beta_3 \gamma_1) x_{1i} + (\beta_2 + \beta_3 \gamma_2) x_{2i} + (\beta_3 \xi_i + u_i) \end{aligned}$$

- (2 points) Explain the residual  $\beta_3 \xi_i + u_i$  is orthogonal to  $(1, x_{1i}, x_{2i})$ . In the long regression, the first-order conditions of the minimization problem will tell us  $u_i$  is orthogonal to  $(1, x_{1i}, x_{2i}, x_{3i})$ . Also in the auxiliary regression  $E[\xi_i] = E[\xi_i x_{1i}] = E[\xi_i x_{2i}] = 0$ . Therefore, the linear combination  $(\beta_3 \xi_i + u_i)$  is orthogonal to  $(1, x_{1i}, x_{2i})$ .
- (2 points) Given that  $(\alpha_0, \alpha_1, \alpha_2)$  is the unique solution to  $\min_a E[(y_i - a_0 - a_1 x_{1i} - a_2 x_{2i})^2]$ , it is the unique solution to the FOCs  $E[\nu_i] = E[\nu_i x_{1i}] = E[\nu_i x_{2i}] = 0$ . Therefore, we have

$$\alpha_2 = \beta_2 + \beta_3 \gamma_2$$

**(b) (10 points)** Suppose that  $\text{cov}(x_{3i}, x_{1i}) = 0$  and  $\text{cov}(x_{3i}, x_{2i}) = 0$ . Does  $\alpha_2 = \beta_2$ ? Explain.

• **Solution: Yes.**

- (4 points if the student points out this fact) If  $\text{cov}(x_{3i}, x_{1i}) = 0$  and  $\text{cov}(x_{3i}, x_{2i}) = 0$ , the auxiliary regression will have  $\gamma_1 = \gamma_2 = 0$ .
- (4 points if there is an explanation for why  $\gamma_1 = \gamma_2 = 0$ .) By Frisch-Waugh Theorem,

$$\gamma_2 = \frac{\text{cov}(x_{3i}, x_{2i} - (\pi_0 + \pi_1 x_{1i}))}{\text{var}(x_{2i} - (\pi_0 + \pi_1 x_{1i}))}$$

where  $x_{2i} = \pi_0 + \pi_1 x_{1i} + \eta_i$  is the population regression of  $x_{1i}$  on  $(1, x_{2i})$ . Given that  $x_{3i}$  is uncorrelated with both  $x_{1i}$  and  $x_{2i}$ , we have  $\gamma_2 = 0$ . The same applies to  $\gamma_1$ .

- (2 points) Using the formula in (a), if  $\gamma_2 = 0$ , we have  $\alpha_2 = \beta_2$ .

**(c) (15 points)** Suppose that

$$\text{cov}(x_{2i}, x_{3i}) = 0, \text{cov}(x_{1i}, x_{3i}) \neq 0, \text{ and } \text{cov}(x_{1i}, x_{2i}) \neq 0$$

Does  $\alpha_2 = \beta_2$ ? Explain.

- Solution: **No** if  $\beta_3 \neq 0$ .

- Revisit the auxiliary regressions

$$\begin{aligned}x_{3i} &= \gamma_0 + \gamma_1 x_{1i} + \gamma_2 x_{2i} + \xi_i \\x_{2i} &= \pi_0 + \pi_1 x_{1i} + \eta_i\end{aligned}$$

in which  $\pi_1 = \frac{\text{cov}(x_{2i}, x_{1i})}{\text{var}(x_{1i})} \neq 0$  as  $x_{1i}$  is correlated with  $x_{2i}$  ( $\text{cov}(x_{1i}, x_{2i}) \neq 0$ ).

- By Frisch-Waugh Theorem,

$$\gamma_2 = \frac{\text{cov}(x_{3i}, \eta_i)}{\text{var}(\eta_i)}$$

$$\begin{aligned}\text{in which } \text{cov}(x_{3i}, \eta_i) &= \text{cov}(x_{3i}, x_{2i} - (\pi_0 + \pi_1 x_{1i})) \\&= \text{cov}(x_{3i}, x_{2i}) - \pi_1 \text{cov}(x_{3i}, x_{1i})\end{aligned}$$

$\text{cov}(x_{3i}, x_{2i}) = 0$  but  $\text{cov}(x_{3i}, x_{1i}) \neq 0$  and  $\pi_1 \neq 0$ . Therefore,  $\gamma_2 \neq 0$ .

- (2 points) Using the formula in (a), if  $\gamma_2 \neq 0$ , we have  $\alpha_2 \neq \beta_2$  as long as  $\beta_3 \neq 0$ . If  $\beta_3 = 0$ , there is no OVB.

### Question 3 (40 points; 30 minutes)

In Lecture 3b we examined the role of education in driving the wage gap between native and immigrants. We considered four OLS regressions:

- (1)  $\hat{y}_i = \hat{\pi}_0 + \hat{\pi}_1 \text{Imm}_i$
- (2)  $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 \text{Educ}_i + \hat{\gamma} \text{Imm}_i$
- (3)  $\hat{y}_i = \hat{\beta}_0^a + \hat{\beta}_1^a \text{Educ}_i$  among natives
- (4)  $\hat{y}_i = \hat{\beta}_0^b + \hat{\beta}_1^b \text{Educ}_i$  among immigrants

where  $y_i$  = log wage of person  $i$ ,  $\text{Imm}_i = 1$  if person  $i$  is an immigrant, and  $\text{Educ}_i$  = years of schooling. Models (1)-(2) are estimated on the pooled sample, including immigrants and natives. Model (3)-(4) are estimated separately on natives, and immigrants respective. Let “a” in (3) denote natives, and “b” in (4) denote immigrants. The table below presents the sample estimates.

|                 | Pooled Model: Fit to<br>Natives and Immigrants |                   | Model for<br>Natives | Model for<br>Immigrants |
|-----------------|------------------------------------------------|-------------------|----------------------|-------------------------|
|                 | (1)                                            | (2)               | (3)                  | (4)                     |
| Constant        | 3.013<br>(0.006)                               | 1.546<br>(0.025)  | 1.365<br>(0.033)     | 1.676<br>(0.035)        |
| Immigrant       | -0.247<br>(0.013)                              | -0.072<br>(0.013) | --                   | --                      |
| Education (yrs) | --                                             | 0.104<br>(0.002)  | 0.117<br>(0.002)     | 0.088<br>(0.002)        |
| MSE             | 0.757                                          | 0.695             | 0.689                | 0.707                   |
| Adj. R-sq       | 0.018                                          | 0.173             | 0.146                | 0.208                   |
| Sample Size     | 19,092                                         | 19,092            | 14,921               | 4,141                   |

Notes: Fit to data for males age 30-45 in March 2012 CPS. Dependent variable is log average hourly wage. Standard errors in parentheses.

Note: question (a)-(b) use information from the regression table, whereas (c) asks for an algebraic proof.

**(a) (10 points)** What is the average log wage gap between immigrants and natives,  $\bar{y}^b - \bar{y}^a$ ? Provide a numeric answer and explain.

• **Solution:**

- (5 points on the number)  $\bar{y}^b - \bar{y}^a = -0.247$  according to column (1). Immigrants earn 25 log points (or percent) less than natives.
- (5 points on explanation) Model (1) is the regression of log wage on a constant and an indicator for immigrant. The coefficient on the Immigrant dummy,  $\hat{\beta}_1 = \bar{y}^b - \bar{y}^a$  is the difference in mean between immigrants and natives.

**(b) (15 points)** On average immigrants have 12 years of schooling ( $\overline{Educ}^b = 12$ ), and natives have 14 years of schooling ( $\overline{Educ}^a = 14$ ). How much can the log wage gap in (a) be explained by differences in education attainment? Provide a numeric answer and explain. *Hint: You may apply Oaxaca decomposition to either the pooled regression (2) or separate models (3)-(4).*

• **Solution: either answer is correct.**

- **(Answer 1)** Provide a decomposition of  $\bar{y}^b - \bar{y}^a$  via estimates in the pooled regression - column (2).

- (5 points) Regression (2) includes a constant and controls for an indicator for Immigrants. It can **fit the mean** of each group:

$$\begin{aligned}\text{native } \bar{y}^a &= \hat{\beta}_0 + \hat{\beta}_1 \overline{Educ}^a \\ \text{immigr } \bar{y}^b &= \hat{\beta}_0 + \hat{\beta}_1 \overline{Educ}^b + \hat{\gamma}_1\end{aligned}$$

- (5 points on the decomposition) The mean gap can be decomposed as follows:

$$\begin{aligned}\bar{y}^b - \bar{y}^a &= \hat{\beta}_0 + \hat{\beta}_1 \overline{Educ}^b + \hat{\gamma}_1 - (\hat{\beta}_0 + \hat{\beta}_1 \overline{Educ}^a) \\ &= \hat{\beta}_1 (\overline{Educ}^b - \overline{Educ}^a) + \hat{\gamma}_1\end{aligned}$$

- (5 points on pointing out the “between” component and providing a numeric answer; rounding error ok)  $\hat{\beta}_1 (\overline{Educ}^b - \overline{Educ}^a)$  represents the part of the wage gap that can be explained by difference in education attainment. Based on the estimate in column (2), this “between” component equals to  $0.104 * (12 - 14) = -0.208$ . That is, the difference in education composition can explain  $0.208/0.247$  roughly 80% of the wage gap between natives and immigrants.
- **(Answer 2)** Use the regressions (3)-(4) estimated separately for immigrants vs. natives:

- (5 points) Regressions (3)-(4) both include a constant and can **fit the mean** of each group:

$$\begin{aligned}\text{native } \bar{y}^a &= \hat{\beta}_0^a + \hat{\beta}_1^a \overline{Educ}^a \\ \text{immigr } \bar{y}^b &= \hat{\beta}_0^b + \hat{\beta}_1^b \overline{Educ}^b\end{aligned}$$

- (5 points on the decomposition) The mean gap can be decomposed as follows:

$$\begin{aligned}\bar{y}^b - \bar{y}^a &= \hat{\beta}_0^b + \hat{\beta}_1^b \overline{Educ}^b - (\hat{\beta}_0^a + \hat{\beta}_1^a \overline{Educ}^a) \\ &= (\hat{\beta}_0^b - \hat{\beta}_0^a) + \underbrace{\hat{\beta}_1^b (\overline{Educ}^b - \overline{Educ}^a)}_{(*)} + (\hat{\beta}_1^b - \hat{\beta}_1^a) \overline{Educ}^a \\ &= (\hat{\beta}_0^b - \hat{\beta}_0^a) + \underbrace{\hat{\beta}_1^a (\overline{Educ}^b - \overline{Educ}^a)}_{(**)} + (\hat{\beta}_1^b - \hat{\beta}_1^a) \overline{Educ}^a\end{aligned}$$

Either the 2nd equation or the 3rd equation works.

- (5 points on pointing out the “between” component and providing a numeric answer; rounding error ok). (\*) or (\*\*) represent the between component, the part of wage gap that can be explained by difference in education attainment.

$$(*) : \hat{\beta}_1^b (\overline{Educ}^b - \overline{Educ}^a) = 0.088 * (12 - 14) = -0.176$$

$$(**) : \hat{\beta}_1^a (\overline{Educ}^b - \overline{Educ}^a) = 0.117 * (12 - 14) = -0.234$$

which suggests the composition effect could explain  $\frac{0.176}{0.247} \approx 71\%$  or  $\frac{0.234}{0.247} \approx 94\%$  of the log wage gap.

(c) (15 points) The pooled estimate of return to education,  $\hat{\beta}_1$  in model (2) is a weighted average of  $\hat{\beta}_1^a$  in model (3) and  $\hat{\beta}_1^b$  in model (4):

$$\hat{\beta}_1 = \omega_a \hat{\beta}_1^a + \omega_b \hat{\beta}_1^b$$

For ease of notation, define  $x_i = Educ_i$ . By Frisch-Waugh Theorem for OLS in sample, we have:

$$\hat{\beta}_1 = \frac{\frac{1}{N} \sum_i y_i \hat{\xi}_i}{\frac{1}{N} \sum_i \hat{\xi}_i^2}$$

where  $\hat{\xi}_i = x_i - \hat{\pi}_0 - \hat{\pi}_1 Imm_i$  is the residual from the auxiliary regression. Define  $N^a$  = number of natives in sample, and  $\hat{p}^a = \frac{N^a}{N}$  = fraction of natives in sample. And the denominator of  $\hat{\beta}_1$  represents the sample variance of the residual,  $\hat{var}(\hat{\xi}_i) = \frac{1}{N} \sum_i \hat{\xi}_i^2$ .

**Question:** Find the weights  $\omega_a, \omega_b$  as functions of  $\hat{p}^a, \hat{var}(\hat{\xi}_i)$ , and group-specific variance  $\hat{var}(x_i|a)$  among natives, and  $\hat{var}(x_i|b)$  among immigrants.

*Hint: Find  $\hat{\pi}_0, \hat{\pi}_1$  in the auxiliary regression. Plug them into the numerator of  $\hat{\beta}_1$  and see how the numerator of  $\hat{\beta}_1$  is related to  $\hat{\beta}_1^a, \hat{\beta}_1^b$ . Note  $\hat{var}(x_i|a) = \frac{1}{N^a} \sum_{i \in a} (x_i - \bar{x}^a)^2$ .*

- **Solution:** In the auxiliary regression,

$$\begin{aligned} \hat{\pi}_0 &= \bar{x}^a, \hat{\pi}_1 = \bar{x}^b - \bar{x}^a \\ \hat{\xi}_i &= x_i - \bar{x}^a - Imm_i \times (\bar{x}^b - \bar{x}^a) \\ &= (1 - Imm_i) \times (x_i - \bar{x}^a) \\ &\quad + Imm_i \times (x_i - \bar{x}^b) \end{aligned}$$

Let's focus on the numerator of  $\hat{\beta}_1$ :

$$\begin{aligned} \frac{1}{N} \sum_i y_i \hat{\xi}_i &= \frac{1}{N} \sum_i y_i ((1 - Imm_i)(x_i - \bar{x}^a) + Imm_i(x_i - \bar{x}^b)) \\ &= \frac{1}{N} \left( \sum_{i \in a \text{ native}} y_i (x_i - \bar{x}^a) + \sum_{i \in b \text{ immig}} y_i (x_i - \bar{x}^b) \right) \\ &= \frac{N^a}{N} \left( \frac{1}{N^a} \sum_{i \in a} y_i (x_i - \bar{x}^a) \right) + \frac{N^b}{N} \left( \frac{1}{N^b} \sum_{i \in b} y_i (x_i - \bar{x}^a) \right) \end{aligned}$$

The separate regressions (3)-(4) are univariate models:

$$\begin{aligned} \hat{\beta}_1^a &= \frac{\frac{1}{N^a} \sum_{i \in a} y_i (x_i - \bar{x}^a)}{\frac{1}{N^a} \sum_{i \in a} (x_i - \bar{x}^a)^2} = \frac{\frac{1}{N^a} \sum_{i \in a} y_i (x_i - \bar{x}^a)}{\hat{var}(x_i|a)} \\ \hat{\beta}_1^b &= \frac{\frac{1}{N^b} \sum_{i \in b} y_i (x_i - \bar{x}^b)}{\hat{var}(x_i|b)} \end{aligned}$$

Therefore,

$$\begin{aligned}
\frac{1}{N} \sum_i y_i \hat{\xi}_i &= \hat{p}^a \text{var}(x_i|a) \hat{\beta}^a + (1 - \hat{p}^a) \text{var}(x_i|b) \hat{\beta}^b \\
\Rightarrow \hat{\beta}_1 &= \frac{\hat{p}^a \text{var}(x_i|a) \hat{\beta}^a + (1 - \hat{p}^a) \text{var}(x_i|b) \hat{\beta}^b}{\text{var}(\xi_i)} \\
&= \underbrace{\frac{\hat{p}^a \text{var}(x_i|a)}{\text{var}(\xi_i)}}_{\omega_a} \times \hat{\beta}^a + \underbrace{\frac{(1 - \hat{p}^a) \text{var}(x_i|b)}{\text{var}(\xi_i)}}_{\omega_b} \times \hat{\beta}^b.
\end{aligned}$$